

Лабораторная №1

1) $X_1 = 1111$, $X_2 = 1001$; $X_1 + X_2 = 2112$; $X_2 - X_1 = -110$

$$\begin{array}{r}
 1111 / 2 \\
 -1110 \\
 \hline
 555 / 2 \\
 1 \quad 554 \\
 \hline
 277 / 2 \\
 1 \quad 276 \\
 \hline
 138 / 2 \\
 1 \quad 138 \\
 \hline
 69 / 2 \\
 0 \quad 68 \\
 \hline
 34 / 2 \\
 1 \quad 34 \\
 \hline
 17 / 2 \\
 0 \quad 16 \\
 \hline
 8 / 2 \\
 1 \quad 8 \\
 \hline
 4 / 2 \\
 0 \quad 4 \\
 \hline
 2 / 2 \\
 0 \quad 2 \\
 \hline
 1 / 2 \\
 0 \quad 1 \\
 \hline
 0
 \end{array}$$

$$1111_{10} = 1000101011_2$$

$$\begin{array}{r}
 1001 / 2 \\
 1000 \\
 \hline
 500 / 2 \\
 1 \quad 500 \\
 \hline
 250 / 2 \\
 0 \quad 250 \\
 \hline
 125 / 2 \\
 0 \quad 124 \\
 \hline
 62 / 2 \\
 1 \quad 62 \\
 \hline
 31 / 2 \\
 0 \quad 30 \\
 \hline
 15 / 2 \\
 1 \quad 14 \\
 \hline
 7 / 2 \\
 1 \quad 6 \\
 \hline
 3 / 2 \\
 1 \quad 2 \\
 \hline
 1 / 2 \\
 1 \quad 1 \\
 \hline
 0
 \end{array}$$

$$1001_{10} = 1111101001_2$$

2)
$$\begin{array}{r}
 1000101011 \\
 0111101001 \\
 \hline
 10000100000
 \end{array}$$

$$64 + 2048 = 2112$$

3) $-1111 = 01000101011$

инверсировав 4 доб. единицы 101110101001

$$\begin{array}{r}
 00111101001 \\
 101110101001 \\
 \hline
 111110010010
 \end{array}$$

Проверка:

$$\begin{array}{r}
 000001101101 \\
 + 1 \\
 \hline
 000001101110 = 110
 \end{array}$$

$$000001101110 = 110$$

$$4) \quad 1111_2 = 457_{16}$$

$$1001_2 = 3E9_{16}$$

$$\begin{array}{r|l}
 1111 & 16 \\
 \hline
 1104 & 69 \quad 16 \\
 \hline
 7 & 64 \quad 4 \\
 \hline
 & 5
 \end{array}$$

$$\begin{array}{r|l}
 1001 & 16 \\
 \hline
 992 & 62 \quad 16 \\
 \hline
 9 & 48 \quad 3 \\
 \hline
 & 14
 \end{array}$$

$$\begin{array}{r}
 5) \quad 457 \\
 \quad 3E9 \\
 \hline
 840_{16} = 2112_{10}
 \end{array}$$

$$\begin{array}{r}
 6) \quad 457 \\
 \quad 3E9 \\
 \hline
 06E_{16} = 110_{10}
 \end{array}$$